CONFAI 2026 · PLAKSHA UNIVERSITY · TRACK 06 — RESPONSIBLE & ETHICAL AI

Refuse to Hallucinate

A deterministic validity firewall for Indian official survey statistics

MoSPI's tools help you
find
the number. This firewall checks whether you are
allowed to say
it — and stops you, with a cited rule, when you are not.

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Poster + live demo · localhost:8001/demo · offline · no API key · no GPU · $0.00 / query

1 The problem

Retrieval over official PDFs is solved. **Governance is not.** Unconstrained RAG over sovereign statistics fails in three ways — observed live in MoSPI's StatsDoc pilot (*archived transcripts, chats 2472–2480*, per `evaluation/statsdoc_audit/`):

- **Inconsistent figures in one answer.** Measured MP MPCE quoted as both 3,158 and 5,589 for urban, and 5,538 vs 4,987 for "All", in a single response (chat 2474).
- **Unguided cross-round arithmetic.** ~40% MPCE growth quoted across the HCES 2011-12 → 2023-24 break (paper → 3-visit CAPI) with no caveat.
- **Retrieval poisoning.** A poisoned document chunk can override guardrails and flip a verdict.

The thesis. You cannot fix a probabilistic system with more probabilistic prompts. The decision — *is this claim allowed?* — must come from **versioned, deterministic rules**, before generation. Models translate; rules judge.

WHERE IT SITS

A drop-in governance layer for journalists, fact-checkers, legislative researchers and statistical agencies — a standalone service/CLI **and** an MCP tool that other agents call as a governance oracle.

Service — `POST /ask`, one JSON endpoint, streaming variant included.

MCP tool — `vidhi.mcp_server`, 4 governance tools, schemas validated.

Agent — `POST /analyze` decomposes a policy question and briefs each claim.

Library — `vidhi.rules.Firewall` imported directly, no service needed.

One answer, end to end — a real response from the running service

"mpce for madhya pradesh"

VERDICT **PASS_WITH_NUMBER** RULE R_OK

3,522 Rs/month

Madhya Pradesh · rural · with imputation — and the slice is not narrowed: the under-specified question returns **every published cell**.

CELL	FIGURE	ROW ID
rural · with imputation	3,522	…rural_with_MadhyaPradesh
urban · with imputation	5,589	…urban_with_MadhyaPradesh
rural · without imputation	3,441	…rural_MadhyaPradesh
urban · without imputation	5,538	…urban_MadhyaPradesh

```
source HCES 2023-24 factsheet, Statement 7
row HCES_2023_24_MPCE_2023-24_rural_with_MadhyaPradesh
verdict hash 7061f3ac… · tier VERIFIED
```

No number is shown without a row id, a PDF page and a hash.

THE RULE VOCABULARY — EVERY REFUSAL IS CITED

A verdict is never bare prose: it carries a stable reason code.

- `R_OK` — verified row matched
- `R_NO_CURATED_ROW` — slice not curated → coverage note, no figure
- `R_NUMERIC_MISMATCH` — asserted number ≠ the store
- `R_COMPARABILITY_*` — survey break → mandatory caveat
- `R_SCOPE_OUT / R_FORECAST` — forecast, causal, political → refused
- `R_PROMPT_INJECTION` — injection refused before any lookup
- `R_METADATA_*` — taxonomy check failed
- `R_LOW_EXTRACTION_CONF` — extraction confidence too low

2 The firewall

1 **Question** — natural language, any register

↓

2 **Typed claim** — translator: models translate, never judge

↓

3 · Firewall — five stages, fixed order, first rule fires

- **Safety-A** — injection · EN / Hindi / Hinglish
- **Scope** — forecast · causal · politics
- **Comparability** — survey breaks
- **Metadata** — taxonomy check
- **Numeric** — verified rows only

↓

4 · SHA-256 freeze — computed **BEFORE** retrieval
`hash = SHA-256(claim + decision + reason + rules)`

↓

5 Retrieve (untrusted document text)

↓

6 **Templated render + citations** — WARN/REFUSE wording is fixed

Safety-B · freeze before retrieval + figure attestation

The verdict hash is locked before any document is read. A planted "ignore the caveat" chunk changes nothing — the hash is byte-identical with or without poisoning. A second hash binds the **number you see** to its cell.

```
frozen_hash = SHA-256(claim ⊕ decision ⊕ sub_code ⊕ reason ⊕ rule_version)
attestation = SHA-256(frozen_hash ⊕ row_id ⊕ value ⊕ unit ⊕ pdf_sha256)
```

Field-guarded cache · no safe threshold

Danger-pair and paraphrase cosine ranges overlap on both measured embedders, so **no similarity threshold is safe**. Reuse only on exact critical-field match.

```
0 collisions across 19 danger pairs · 1708 verified rows (55 touched by the gol
den set)
```

Generalised, not hard-coded

The claim translator resolves **every** state/UT the store carries, in any spelling — "mp", "up", "m.p.", "orissa", "maharastra" — and answers one state or several at once. A slice the question leaves open is answered with **every published cell**, each with its own row id: complete, never silently narrowed.

```
"MPCE in Madhya Pradesh, UP, Maharashtra" → 12 cells · 12 row ids
```

FIRST RULE FIRES — TWO REAL TRACES

STAGE	QUESTION	OUTCOME
1 Safety-A	"Compare average MPCE in 2011-12 vs 2023-24."	pass — no injection signal
2 Scope		pass — descriptive
3 Comparability		**WARN** R_COMPARABILITY_HCES_METHOD — 68th-round paper vs 3-visit CAPI
1 Safety-A	"Will unemployment fall in 2027?"	pass — no injection signal
2 Scope		**REFUSE** R_FORECAST — no such round exists

A stage that fires ends the adjudication: stages 4–5 are never reached in either trace. The order is the guarantee.

What stops the three attacks

Prompt injection — Safety-A refuses before any lookup. **Poisoned chunk** — the hash is already frozen; the chunk changes prose, never the verdict. **Cache poisoning** — exact critical-field match only.

3 Results

- **0.000** — hallucination rate — 0 of 36 must-abstain questions answered wrongly
- **1.000** — abstention recall — every must-abstain question refused/warned
- **100**/100 — verdict accuracy [Wilson 0.963–1.000]
- **60**/60 — numeric exact match [0.940–1.000]
- **0**/19 — danger-pair cache collisions
- **1708**/1708 — verified rows with provenance
- **1.000** — Safety-A P/R/F1 · n=28 (EN · hi · hi-en)
- **$0.00** — marginal cost · p50 < 200 ms, CPU-only

VIDHI VS STATSDOC — HONEST COMPARISON

CHECK	STATSDOC (ARCHIVED)	VIDHI (MEASURED)
Definition	"Monthly…" (got lucky)	statutory, pinned
Consistency	3,158 vs 5,589 urban	one row id per figure
Cross-round math	~40% growth, no caveat	halts with R_COMPARABILITY_HCES_METHOD
Poisoned context	overrides verdict	SHA-256 frozen pre-retrieval
Cost / latency	~$0.02–0.05 · 3.5–8 s *(est.)*	$0.00 · p50 < 200 ms

StatsDoc rows are transcribed from archived live transcripts (chats 2472–2480, same nine prompts); Vidhi rows are measured offline. One prompt set, one session — an observation, not a benchmark.

HONEST LIMITS

- **Coverage is the limit, not logic** — 1,708 verified rows: the PLFS Highlights battery (11 rounds × usual + CWS × LFPR/WPR/UR × 15+/all-ages × 9 sector-sex cells of both annual reports), the TUS 2024 battery (9 ICATUS divisions × gender/sector/age × participation/per-participant/per-person), and the HCES 2023-24 state + trend columns. State-wise MPCE covers **all 36 states/UTs** in any spelling; an open slice returns **every published cell**, never one narrowed figure. Uncovered slices (state-level labour, non-badge rounds) return `WARN_NO_DATA` + a coverage note, never a guess.
- Heuristic extraction breaks on exotic phrasing; an LLM translator port exists (`VIDHI_USE_LLM=1`), unmeasured.
- No load test yet; eSankhyiki live-API path inherits third-party cert-verification weakness (query path is offline).

WHAT YOU CAN INSPECT

Rules, store, claim translator, service, MCP server — `vidhi/`; 1,708 rows, each sha256-bound to its source PDF.

Archived StatsDoc transcripts (chats 2472–2480) + access log + comparison — `evaluation/statsdoc_audit/`: reproducible, not anecdotal.

The four official MoSPI PDFs — `data/raw/`, committed, so every figure can be re-checked at the page.

117 tests — golden set, danger pairs, 28-case multilingual red-team, Safety-B hash invariant.

RUNTIME

p50 150–190 ms warm, CPU-only — the deterministic adjudication itself is sub-millisecond.

Cold start 13–43 s — one-time model load, paid at startup, never per query.

$0.00 marginal cost — no API key, no GPU, no network on the query path.

Laptop, server or small container — the same image runs the demo and the service.

Six guarantees — the soul of the system

1. Rules judge, models translate
2. Freeze before retrieval
3. No cosine threshold is safe
4. No PASS without provenance
5. Ambiguity is never contradiction
6. Honest coverage — never a guess

REPRODUCE EVERYTHING (OFFLINE)

```
# no API key · no GPU · no network
pip install -r requirements.txt
python -m pytest tests/ -q        # 117 passed · every store row reachable
python vidhi/eval.py               # 100 golden + 19 danger + 28 red-team + Safety-B
python -m uvicorn vidhi.service:app --port 8001  # → /demo
```

TRY THE LIVE DEMO

localhost:8001/demo — ask a PLFS / HCES / TUS / MIS question. Watch the verdict freeze, the citations resolve, and every figure name its source row.

MCP: `python -m vidhi.mcp_server` (4 governance tools) · **Agent:** `POST /analyze` for a policy briefing.

THE QUESTION THIS WORK WON'T HALLUCINATE

Ask the hard question — expect `WARN_NO_DATA` + a coverage note, never a guess. Inject in English, Hindi or Hinglish and watch Safety-A refuse with a cited rule.

Questions and red-team attempts welcome.